

Secure Shell (SSH)

- Topics covered:
 - Console Port Security
 - Layer 2 Switch Management IP
 - Telnet
 - SSH

Console Port Security

Why Console Port Security Imp?

By default, a Cisco IOS device does not require a password when accessed through the console port. A new device can be connected to with a console cable and configured immediately through the CLI.

To secure console access, configure a password on the console line. The console line contains all settings related to console-port connections.

Configuring a Console Password

```
R1(config)#line console 0
R1(config-line)#password ccna
R1(config-line)#login
R1(config-line)#end
R1#exit

R1 con0 is now available
Press RETURN to get started.

User Access Verification
Password:
R1>
```

There is only a single console line, so the number is always 0.

Configure the console line's password.

Tell the device to require a user to enter the configured password to access the CLI via the console port.

The password isn't displayed as you type it.

Explanation:

line console 0

1. Enters console line configuration mode.
2. There is only one console line, so the number is always 0.
3. Because there is only a single console line, only one console connection can exist at a time.

password ccna → Configures the console line password.

Login → Requires users to enter the configured password before accessing the CLI.

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After reconnecting through the console port, the device prompts for the password, the password is not displayed while it is being typed.

Using Local User Accounts Instead

Instead of using a password configured directly on the console line, the device can require users to authenticate with a configured username and password.

CLI Output

```
R1(config)#username jeremy secret ccnp
R1(config)#line console 0
R1(config-line)#login local
R1(config-line)#end
R1#exit

R1 con0 is now available

Press RETURN to get started.

User Access Verification
Username: jeremy
Password:
R1>
```

Tell the device to require a user to login using one of the configured usernames on the device.

Explanation:

username jeremy secret ccnp

1. Creates a local user account named jeremy.
2. Configures a secret password of ccnp.

login local

1. Configures the console line to use locally configured usernames.
2. Users must authenticate with a username and password.

Even though the previously configured console password remains in the configuration, it is no longer used because login local changes the authentication method.

Current Console Line Configuration:

```
line con 0
exec-timeout 3 30
password ccna
logging synchronous
login local
```

Log the user out after 3 minutes and 30 seconds of inactivity.

exec-timeout 3 30

1. Logs the user out after 3 minutes and 30 seconds of inactivity.
2. This helps prevent unauthorized access if a user leaves a console session unattended.

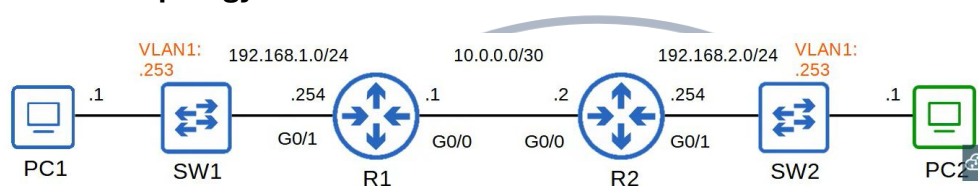
Layer 2 Switch Management IP

Management IP Address on Layer 2 Switches

Routers and Layer 3 switches can be managed remotely through their IP addresses.

Layer 2 switches do not perform packet routing and do not maintain routing tables. However, they can still be assigned a management IP address through an SVI (Switch Virtual Interface) so that administrators can connect remotely using Telnet or SSH.

Network Topology



PC2 is used by the network administrator and must be able to remotely manage devices throughout the network.

Configuring the Management IP

```
SW1(config)#interface vlan1
SW1(config-if)#ip address 192.168.1.253 255.255.255.0
SW1(config-if)#no shutdown
SW1(config-if)#exit
```

Configure the IP address on the SVI in the same way as on a multilayer switch. Enable the interface if necessary.

```
SW1(config)#ip default-gateway 192.168.1.254
```

Configure the switch's default gateway. In this case, PC2 isn't in the same LAN as SW1. If SW1 doesn't have a default gateway, it can't communicate with PC2.

Explanation:

interface vlan1 → Enters the SVI configuration.

ip address 192.168.1.253 255.255.255.0 → Assigns the management IP address.

no shutdown → Enables the interface if it is administratively down.

ip default-gateway 192.168.1.254

1. Configures the switch's default gateway.
2. This is required for communication outside the local LAN.

Without a default gateway, SW1 cannot communicate with PC2 because PC2 is located in a different network.

Telnet

What is Telnet?

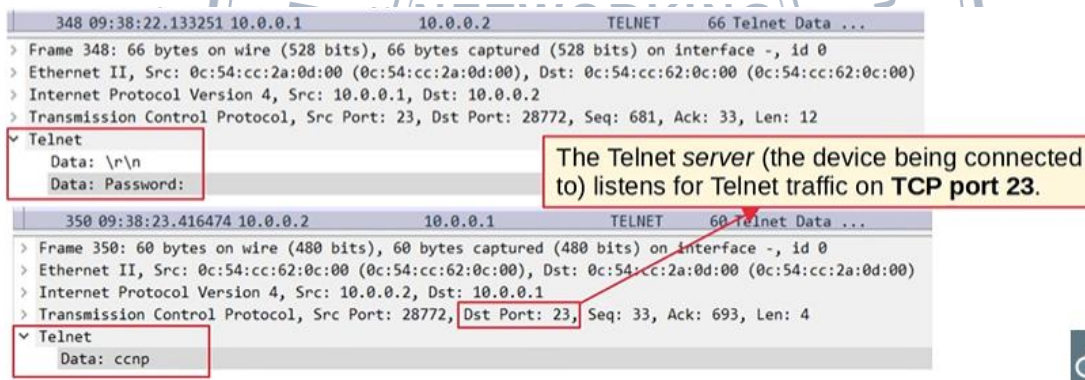
Telnet (Teletype Network) is a protocol used to remotely access the CLI of another device.

Instead of connecting directly with a console cable, administrators can manage devices across the network.

- Developed in 1969.
- Largely replaced by SSH.
- Sends all data in plain text.
- Provides no encryption.

Because traffic is unencrypted, anyone who captures the packets can read the transmitted information.

Telnet Traffic Example (Packet Capture Representation)



Explanation:

The login prompt is transmitted in plain text.

The password entered by the user (ccnp) is also transmitted in plain text, making it visible to anyone who captures the traffic.

This includes:

1. Usernames
2. Passwords
3. Subsequent session traffic

Because of this lack of security, SSH is preferred over Telnet.

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Telnet Port Number

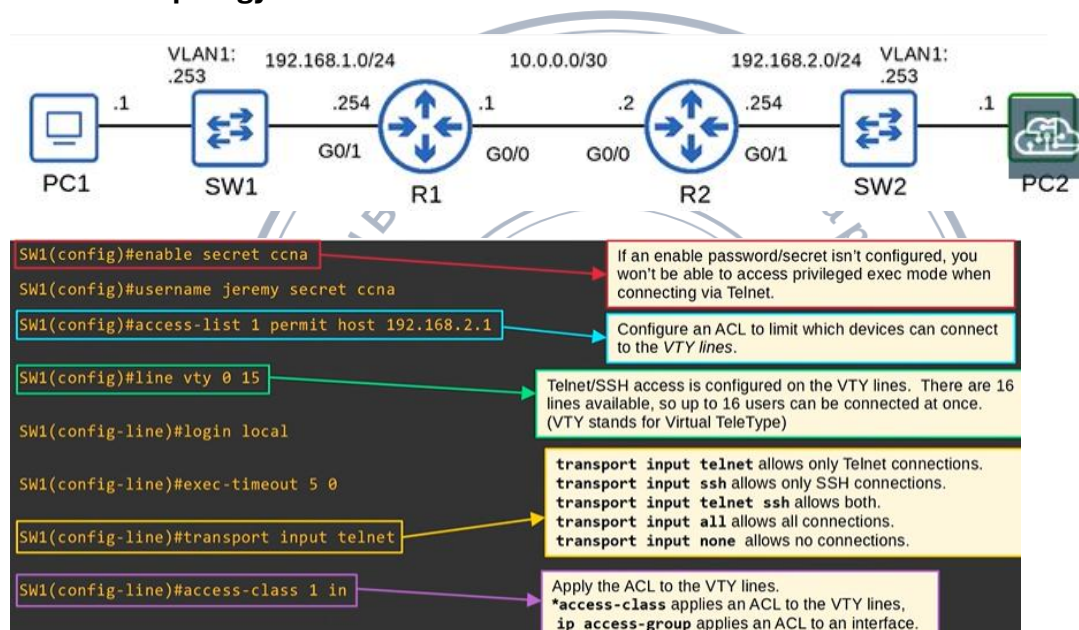
1. The Telnet server listens on TCP port **23**.
2. The Telnet client is the device initiating the connection.
3. The Telnet server is the device receiving the connection.

Remember Telnet = TCP Port 23

Configuring Telnet Access

Let's Configure Telnet

Network Topology



Explanation:

enable secret ccna → Required to access privileged EXEC mode after connecting through Telnet.

username jeremy secret ccna → Creates a local user account.

access-list 1 permit host 192.168.2.1 → Creates an ACL that permits only PC2.

line vty 0 15

1. Configures all VTY lines.
2. VTY stands for **Virtual TeleType**.
3. There are 16 VTY lines, allowing up to 16 simultaneous remote connections.

login local → Requires authentication with local user accounts.

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exec-timeout 5 0 → Disconnects inactive users after 5 minutes.

transport input telnet → Allows only Telnet connections.

Verifying Telnet Access

```
R2#ping 192.168.1.253
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.253, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 10/11/16 ms

R2#telnet 192.168.1.253
Trying 192.168.1.253 ...
% Connection refused by remote host
```

Explanation:

1. R2 can successfully ping SW1.
2. Telnet access is denied because the ACL on the VTY lines only permits PC2.

Successful Connection from PC2:

```
C:\Users\user>telnet 192.168.1.253
Connecting To 192.168.0.1...

User Access Verification

Username: jeremy
Password:
SW1>
```

PC2 is permitted by the VTY ACL, so the Telnet connection succeeds.

VTY Line Display in Running Configuration

```
line vty 0 4
 access-class 1 in
 exec-timeout 5 0
 login local
 transport input telnet
line vty 5 15
 access-class 1 in
 exec-timeout 5 0
 login local
 transport input telnet
```

Although all 16 VTY lines were configured together using line vty 0 15, they appear as two separate sections in the running configuration.

1. Lines 0–4 are displayed first.
2. Lines 5–15 are displayed separately.

This is simply how the configuration is displayed and does not affect operation.

SSH (Secure Shell)

What is SSH?

SSH (**Secure Shell**) was developed in 1995 to replace less secure protocols such as Telnet.

A **shell** is a computer program that exposes operating system services to a human user or another program. Accessing a device's CLI means interacting with a shell.

SSHv2, a major revision of SSHv1, was released in 2006. Version 2 is more secure and should be used whenever possible.

If a device supports both SSH version 1 and version 2, it reports **version 1.99**.

1. SSH 1.99 is **not** an actual SSH version.
2. It simply indicates support for both SSHv1 and SSHv2.

SSH provides security features such as:

- Data encryption
- Authentication

Unlike Telnet, captured SSH traffic cannot be read without the proper cryptographic keys.

SSH Encryption Example (Packet Capture Representation)



Explanation:

The SSH packet contents appear as encrypted, seemingly random characters.

Only the SSH client and SSH server possess the keys required to decrypt the data. Even if the packet is captured while travelling across the network, its contents cannot be read.

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SSH Port Number

The SSH server listens for SSH traffic on **TCP port 22**.

The SSH server (the device being connected to) listens for SSH traffic on **TCP port 22**.

SSH -> TCP 22

Telnet -> TCP 23

Verifying SSH Support

Before configuring SSH, verify that the IOS image supports it.

```
SW1#show version
Cisco IOS Software, vios_12 Software (vios_12-ADVENTERPRISEK9-M), Version 15.2(4.0.55)E, TEST
ENGINEERING ESTG WEEKLY BUILD, synced to END_OF_FLO_ISP
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2015 by Cisco Systems, Inc.
Compiled Tue 28-Jul-15 18:52 by sasyamal
```

Explanation:

The IOS image name contains K9.

- IOS images that support SSH contain K9 in the image name.

Cisco also distributes NPE (No Payload Encryption) IOS images to countries with encryption restrictions.

- NPE IOS images do not support cryptographic features such as SSH.

```
SW1#show ip ssh
SSH Disabled - version 1.99
%Please create RSA keys to enable SSH (and of atleast 768 bits for SSH v2).
Authentication methods:publickey,keyboard-interactive,password
Authentication Publickey Algorithms:x509v3-ssh-rsa,ssh-rsa
Hostkey Algorithms:x509v3-ssh-rsa,ssh-rsa
Encryption Algorithms:aes128-ctr,aes192-ctr,aes256-ctr,aes128-cbc,3des-cbc,aes192-cbc,aes256-cbc
MAC Algorithms:hmac-sha1,hmac-sha1-96
Authentication timeout: 120 secs; Authentication retries: 3
Minimum expected Diffie Hellman key size : 1024 bits
IOS Keys in SECSH format(ssh-rsa, base64 encoded): NONE
```

Explanation:

1. SSH is supported.
2. SSH is currently disabled.
3. Version 1.99 indicates support for both SSHv1 and SSHv2.
4. RSA keys must be created before SSH can be enabled.

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Generating RSA Keys

RSA keys are required for:

1. Data encryption
2. Data decryption
3. Authentication
4. Other SSH security functions

The device's **FQDN (Fully Qualified Domain Name)** is used when naming RSA keys.

FQDN = Hostname + DNS Domain Name

```
SW1(config)#ip domain name jeremysitlab.com
SW1(config)#crypto key generate rsa
The name for the keys will be: SW1.jeremysitlab.com
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.
How many bits in the modulus [512]: 2048
% Generating 2048 bit RSA keys, keys will be non-exportable...
[OK] (elapsed time was 1 seconds)
SW1(config)#
*Feb 21 04:22:35.778: %SSH-5-ENABLED: SSH 1.99 has been enabled
```

The **FQDN** of the device is used to name the RSA keys.
FQDN = Fully Qualified Domain Name (host name + domain name)

Generate the RSA keys.
crypto key generate rsa modulus Length
is an alternate method.
*length must be 768 bits or greater for SSHv2

Explanation:

ip domain name jeremysitlab.com → Configures the DNS domain name.

crypto key generate rsa → Generates RSA keys.

The generated key name uses the device FQDN: SW1.jeremysitlab.com

A modulus size must be selected.

In this example: 2048 bits was used.

An alternate command is: crypto key generate rsa modulus length where the key length is specified directly.

Key Length Requirement

For SSHv2: RSA key length $\geq$ 768 bits

Larger keys provide greater security but require more time to generate and use.

Verifying SSH After RSA Generation

```
SW1(config)#do show ip ssh
SSH Enabled - version 1.99
Authentication methods:publickey,keyboard-interactive,password
Encryption Algorithms:aes128-ctr,aes192-ctr,aes256-ctr,aes128-cbc,3des-cbc,aes192-cbc,aes256-cbc
MAC Algorithms:hmac-sha1,hmac-sha1-96
Authentication timeout: 120 secs; Authentication retries: 3
Minimum expected Diffie Hellman key size : 1024 bits
IOS Keys in SECSH format(ssh-rsa, base64 encoded): SW1.jeremysitlab.com
[output omitted]
```

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Explanation:

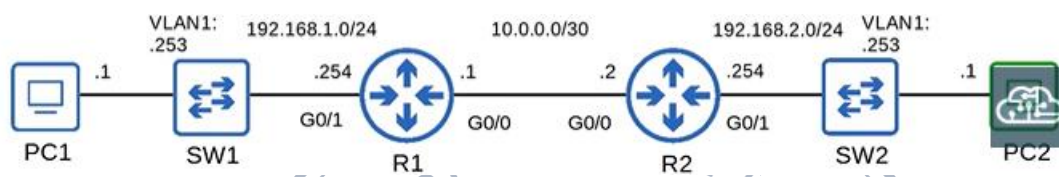
After RSA keys are generated:

1. A Syslog message indicates SSH has been enabled.
2. show ip ssh confirms SSH operation.

SSH Configuration

Let's Configure SSH and see how it's work.

Network Topology



Configuration Commands

```
SW1(config)#enable secret ccna
SW1(config)#username jeremy secret ccna
SW1(config)#access-list 1 permit host 192.168.2.1
SW1(config)#ip ssh version 2
SW1(config)#line vty 0 15
SW1(config-line)#login local
SW1(config-line)#exec-timeout 5 0
SW1(config-line)#transport input ssh
SW1(config-line)#access-class 1 in
```

(optional, but recommended) Restrict SSH to version 2 only.

Configure all VTY lines, just like Telnet.

Enable local user authentication.
*you cannot use **login** for SSH, only **login local**.

(optional, but recommended) Configure the exec timeout.

Best practice is to limit VTY line connections to SSH only.

(optional, but recommended) Apply the ACL to restrict VTY line connections.

Explanation:

enable secret ccna → Configures the enable secret.

username jeremy secret ccna → Creates a local user account.

access-list 1 permit host 192.168.2.1 → Restricts access to PC2.

ip ssh version 2

1. Restricts SSH to version 2 only.
2. Optional but recommended.

line vty 0 15 → Configures all VTY lines.

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login local

1. Enables local user authentication.
2. SSH requires usernames.
3. login cannot be used for SSH.

exec-timeout 5 0 → Configures the inactivity timeout.

transport input ssh

1. Allows only SSH connections.
2. Disables Telnet access.

access-class 1 in

1. Applies the ACL to VTY connections.
2. Optional but recommended.

SSH Configuration Summary

SSH configuration should be performed in the following order.

1. Configure a Hostname

```
Router(config)#crypto key generate rsa
% Please define a hostname other than Router.
```

Explanation:

RSA keys cannot be generated while the device uses the default hostname.

Example: Router(config)#hostname R2

2. Configure a Domain Name

```
R2(config)#crypto key generate rsa
% Please define a domain-name first.
```

Explanation:

A domain name must be configured before RSA key generation.

Example: R2(config)#ip domain name jeremysitlab.com

3. Generate the RSA Key Pair

```
R2(config)#crypto key generate rsa
The name for the keys will be: R2.jeremysitlab.com
[output omitted]
```

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Explanation:

Once both hostname and domain name exist, RSA key generation succeeds.

Remember: Hostname + Domain Name = RSA Key Generation Available

4. Configure Authentication

Configure:

1. Enable password/secret
2. Local username and password

The order of this step is not important.

5. Enable SSH Version 2

`ip ssh version 2`

Not required, but recommended.

6. Configure the VTY Lines

The most important setting is:

`transport input ssh`

Additional VTY configurations can also be applied:

1. `login local`
2. `exec-timeout`
3. `access-class`

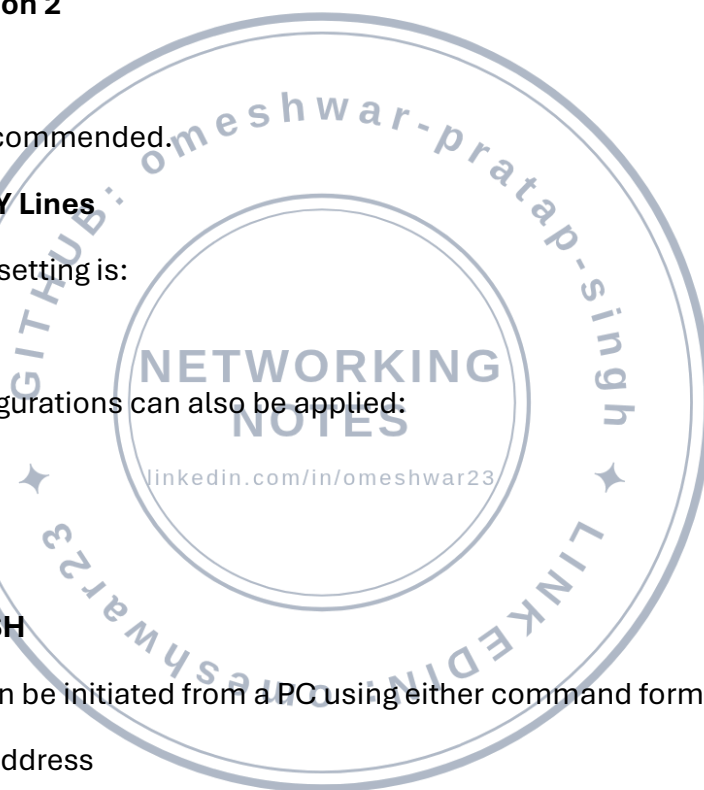
Connecting with SSH

SSH connections can be initiated from a PC using either command format:

`ssh -l username ip-address`

or

`ssh username@ip-address`



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Command Summary

```
SW1# show version
SW1# show ip ssh
SW1(config)# ip default-gateway ip-address
SW1(config)# line con 0
SW1(config)# line vty 0 15
SW1(config)# crypto key generate rsa
SW1(config)# ip ssh version 2
SW1(config-line)# login [local]
SW1(config-line)# transport input [protocols | all | none]
SW1(config-line)# exec-timeout minutes seconds
SW1(config-line)# access-class acl in
```

```
> telnet ip-address
> ssh -l username ip-address
> ssh username@ip-address
```

